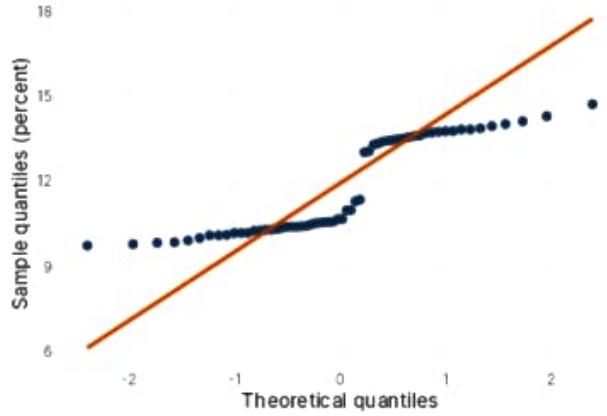
R.No: 4 Reg. No: 192424243 Name: B PALLAVI

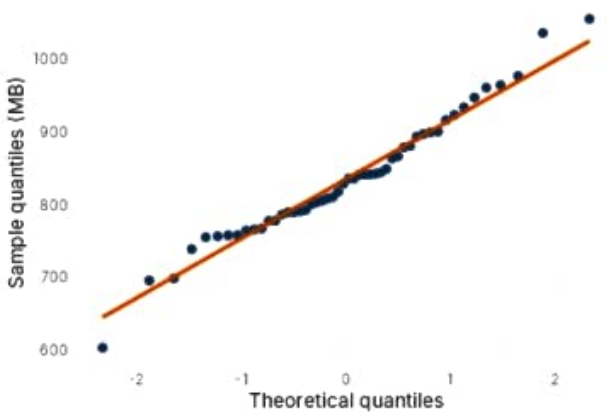
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: battery drain per hour (percent), n = 60



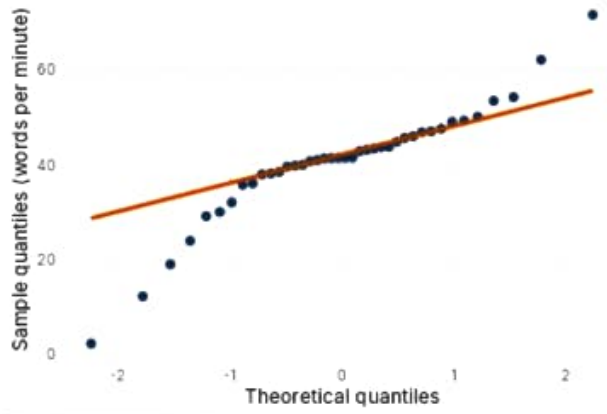
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: mobile data used per day (MB), n = 50



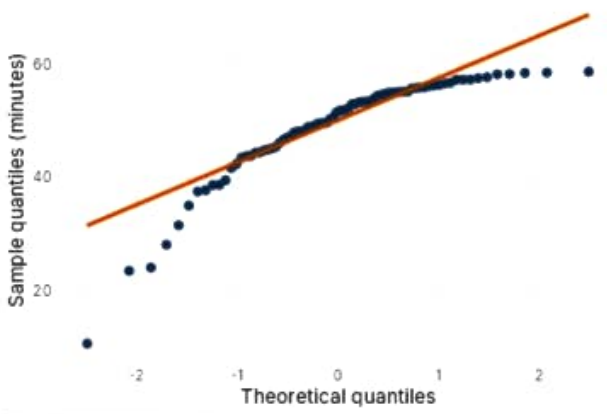
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: typing speed (words per minute), n = 40



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: commute time (minutes), n = 80



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Activity-5

Chart-1

1. Normal? No.
2. Shape : Bimodal.
3. Evidence in the plot.

- The Q-Q plot does not follow the straight reference line. The sample quantiles show two distinct groups or clusters.

Chart-2.

1. Normal ? No
2. Shape : Right-Skewed
3. Evidence in the plot

The upper tail points curve above the reference line, while the lower-tail points fall below it. A few unusually high values are present.

Chart -2:

1. Normal ? No.
2. Shape : Heavy-tailed
3. Evidence : The upper tail points curve above the reference line at both ends, especially in the upper tail, where points lie above the line.

Chart-4

1. Normal? No

2. Shape: left skewed

3. Evidence: The lower tail points fall well below the reference line, while the upper tail points flatten below the line, indicating a lower tail.

Part B

B_1. Chart 3 departure and remedy.

Departure: Right-skewed distribution.

Remedy: Apply a log transformation before using mean and standard deviation methods.

B_2: Fill in the blanks.

```
ggplot(df, aes(sample = Value)) +
    stat_qq() +
    stat_qq_line();
```